Yukawa screening derivation of the bond-valence rule

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The bond-valence model is a standard way to estimate bond strengths in crystals, but its exponential dependence on bond length has lacked a derivation from a specific physical interaction. We show that this form emerges as the leading-order limit of screened Coulomb electrostatics and that the fitted bond-valence softness can be interpreted in terms of an electronic screening length. This turns bond valence from an empirical fitting rule into a transferable descriptor of local screened charge response across coordination environments. The resulting theory predicts how the bond-valence parameters should vary with ionic charge and coordination number, and that prediction agrees with 150 fitted relationships from 94 cation–oxygen species, including 68 in fourfold coordination and 82 in sixfold coordination. The screening length also sets a valence-dependent cation–oxygen contact distance that reproduces tabulated crystal radii, with an effective oxygen size that grows by about 0.09 Å per unit of cation charge. A comparison with first-principles charge densities shows that the bond-valence shell radius tracks the electronic screening cloud with coefficients of determination of 0.9998 for ten alkali and alkaline-earth oxides and 0.967 for 21 other binary oxides whose nearest-neighbor environments match the theory's assumptions. The widely used bond-valence model is thus the leading-order expression of screened electrostatics in ionic solids.

Pauling's observation that the strength of a bond reaching a coordinating anion approximates the cation charge divided by the coordination number [1] has been an organizing principle of crystal chemistry for nearly a century. Brown and Altermatt [2] cast this idea as the bond-valence model: the empirical bond strength between a cation i and anion j separated by R_{ij} is

$$S_{ij} = \exp\left(\frac{R_0 - R_{ij}}{B}\right), \tag{1}$$

where R_0 and B are parameters fitted to crystallographic data and the valence-sum rule $\sum_j S_{ij} = z_i$ (formal charge) is enforced. The model is used routinely for structure validation [3, 4] and increasingly as a structural descriptor in machine-learning models [5–7].

Several lines of work have sought a physical foundation for the bond-valence exponential. Preiser *et al.* [8] showed that bond valences closely track the electrostatic flux linking each cation–anion pair, recasting the model as localized electrostatics constrained by Gauss's law. Brown developed this flux picture into a broader framework in which bonds act as localized capacitors [3, 4]. In these treatments the exponential distance dependence is postulated rather than derived from a specific potential, so the softness B remains an empirical length with no associated interaction. In practice the near-degeneracy of jointly fitted (R_0, B) is well documented [9], and it has been suppressed by fixing $B \simeq 0.37$ Å [2]. Adams and co-workers connected B to Morse-potential curvature and showed that transferable parameter sets require non-universal B [10, 11].

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The (R_0, B) correlation itself has not been treated as physical content of the theory, but here we show that a Yukawa expansion obtains the exponential as the leading term of a specific screened potential, ties B to a measurable screening length, predicts the B–R_0 correlation with no fitted constants, and connects the fitted parameters to electronic structure. In what follows, we extend Brown's postulate that screened flux partitions among coordinating anions, from which isotropic Yukawa weight and its linearization are the two controlled approximations. We show that partition-thermodynamic relations emerge as exact identities of Eq. (1). Modern first-principles methods that compute screened local Coulomb interactions directly from electronic structure [12, 13] are shown to correlate with the Yukawa screening length, establishing a physical foundation for the bond-valence model.

Screened-flux derivation. A cation with charge $Q_i = z_i e$ emits electrostatic flux Q_i/ϵ_0. In a polarizable medium the intervening electrons screen this flux, and the standard static description of screening replaces the Coulomb potential by the Yukawa form $\phi(r) \propto e^{-r/\lambda}/r$, with λ the screening length [14–17]. Applying Gauss's law to this potential, the flux threading a sphere of radius R is $(Q_i/\epsilon_0)\, e^{-R/\lambda}(1 + R/\lambda)$, so the signed flux amplitude captured by bond i–j at length R_{ij} is $\widetilde{\Phi}_{ij} \propto e^{-R_{ij}/\lambda}(1 + R_{ij}/\lambda)$. Preiser *et al.* [8] partitioned flux according to Gauss's law such that each bond carries a valence proportional to the electrostatic flux it subtends. We retain the same correspondence, but for the screened field.

Adopting this single-λ Yukawa weight for every bond of the shell is the physical ansatz of the derivation. Bond valence is a nonnegative partition of the cation valence among its anion neighbors, and two conditions fix that partition. The valence-sum rule $\sum_j S_{ij} = z_i$ sets the total. Screening attenuates the flux, and the sum rule renormalizes the total back to the formal charge, so the screened field determines only the relative apportionment among bonds. Preserving that apportionment means that ratios of bond valences equal ratios of the captured screened-flux magnitudes,

$$\frac{S_{ij}}{S_{ik}} = \frac{w_{ij}}{w_{ik}}, \qquad w_{ij} := |\widetilde{\Phi}_{ij}|.$$

This guarantees the non-negativity that Eq. (1) requires, imports the distance dependence of the screened interaction into the partition, and introduces a single screening parameter, λ, to scale the distance dependence. Together the two conditions uniquely give

$$S_{ij} = z_i \, \frac{w_{ij}}{\sum_k w_{ik}}. \tag{2}$$

To linearize, we expand $\ln w_{ij}$ about a reference bond length R' chosen within the coordination shell, giving

$$\ln w_{ij} = C_i - \frac{R'}{\lambda(\lambda + R')}\,(R_{ij} - R') + \delta_{ij}, \tag{3}$$

where C_i is independent of bond index j and $\delta_{ij} = O[(R_{ij} - R')^2/(\lambda + R')^2]$. Exponentiating the linear term and renormalizing therefore reproduces Eq. (1) at $R_{ij} = R'$ with only quadratic log-weight error away from R'. Define the leading-order Yukawa exponent denominator

$$B = \frac{\lambda(\lambda + R')}{R'}. \tag{4}$$

This B is the first-order exponent denominator produced by the Taylor expansion of the Yukawa weight. The screening length λ is the primary physical variable, and the sign of λ distinguishes screening from anti-screening. For $\lambda > 0$ the Yukawa weight decays with bond length, so flux reaching the more distant anions is attenuated and shorter bonds capture the larger share of the valence. For $-R' < \lambda < 0$ the weight instead grows with bond length, and longer bonds carry the larger share. Fits return this anti-screening sign for 16% of the coordination shells in the corpus analyzed below, but roughly half of those are nearly degenerate shells whose fitted B is barely constrained (Supplemental Material [18]). The chemically systematic cases are square-planar and linear late transition metals (Pd^{2+}, Au^{3+}, linear O–Cu–O) and large, soft cations in high, irregular coordination, where directional bonding or ambiguous coordination produces anisotropic flux capture outside the isotropic Yukawa assumption.

Parameter-free slope. In the uniform-bond limit, where $R_{ij} = \bar{R}$ for all j, the valence-sum rule gives $R_0 = \bar{R} + B\ln(z/n)$. Differentiating:

$$\beta := \frac{dB}{dR_0} = \frac{1}{\ln(z/n)}. \tag{5}$$

The slope of the B–R_0 correlation at fixed coordination number n is determined entirely by the charge-to-coordination ratio z/n, with a pole at $z = n$, with no fitting parameters. Figure 1 shows that Eq. (5) reproduces the fitted slopes for 94 cation–oxygen species at $n = 4$ and $n = 6$ (68 at $n = 4$, 82 at $n = 6$, 150 slopes in total). The weighted R^2 is 0.986, with mean absolute error 0.15 under the weighting scheme defined in the Supplemental Material [18]. Each fitted slope is the regression of one species' per-structure (R_0, B) fits at fixed n, constructed as described in the empirical validation below. The slopes carry bootstrap standard errors, with median relative standard error 0.2% across the 150 slopes and 90th percentile 4%, so the scatter at fixed (z, n)—most visible near $z = 3$—reflects real differences among species rather than fit uncertainty (per-species slopes with error bars in the Supplemental Material [18]).

Origin of the near-pole spread. The spread in Fig. 1 concentrates in the charge classes nearest each pole, and the pole form itself sets that pattern. At leading order $1/\beta = \ln(z/n)$, and the second-order terms δ_{ij} of Eq. (3) perturb this inverse slope, so a species-level perturbation u shifts the fitted slope by $-\beta^2 u$. Fixed chemical scatter is thus amplified quadratically toward the pole, and across the charge classes of Fig. 1 the per-class spread indeed grows as $|\beta|^{1.8}$. The $z = 3$ class at $n = 4$ is the clearest case. The same cations at $n = 6$ match Eq. (5) to a median $|\Delta\beta| = 0.03$, so chemical diversity alone is not the origin. The largest deviations belong to species for which fourfold oxygen coordination is rare, chiefly the trivalent lanthanides and Y (median 17 structures per fit), whereas the well-populated trivalents Ga^{3+} (209 structures), B^{3+} (214), and Fe^{3+} (160) lie on the curve. The deviant slopes are almost all shallower than predicted, consistent with δ_{ij} attenuating the leading-order link in sparse, atypical shells, and the structure-count weighting downweights such shells on physical grounds.

Screening collapse at $z = n$. What degenerates at $z = n$ is the leading-order link between the fitted parameters. The B dependence of R_0 vanishes as $z \to n$, so the intercept ceases to constrain the softness. B is then fixed only by the second-order spread terms δ_{ij} in Eq. (3). The branch change across the pole is carried by the sign reversal of the slope $\beta = 1/\ln(z/n)$, negative for $z < n$ and positive for $z > n$. At $z = n$ the sum rule fixes only the mean bond valence at unity, but it does not force $S_{ij} = 1$ bond by bond. For a truly degenerate shell—all bond lengths equal by symmetry, as for Si^{4+} in zircon ($ZrSiO_4$), whose four Si–O bonds are symmetry-equivalent—Eq. (1) assigns every bond the same valence, and a single structure supplies only the uniform-limit relation $R_0 = \bar{R} + B\ln(z/n)$. B is unconstrained, and at $z = n$ the relation collapses to $R_0 = \bar{R}$ at any B. For a nearly degenerate shell the bond lengths differ, so the S_{ij} must differ and the sum rule is satisfied at finite B. α-quartz is a common example, with two experimental Si–O distances differing by about 0.01 Å that finite B re-

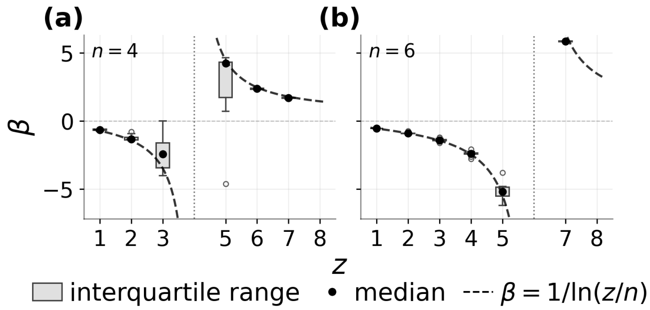

FIG. 1. Fitted slope β vs. formal charge z for 94 cation–oxygen species at (a) $n = 4$ (68 cations) and (b) $n = 6$ (82 cations). 56 cations contribute at both coordinations, giving 150 fitted slopes in total. Boxes span the interquartile range of the per-species slopes at each z, black points mark the median, whiskers extend to the most extreme slopes within 1.5× the interquartile range, and open circles mark slopes beyond the whiskers. Dashed curves: $\beta = 1/\ln(z/n)$. The pole at $z = n$ separates the negative-slope ($z < n$) and positive-slope ($z > n$) branches of Eq. (5). Species at $z = n$ are omitted. Per-species slopes are resolved in the Supplemental Material [18].

solves. Both cases are common. Of the 7,565 shells with $z = n$ in our corpus, 959 are degenerate within 10^{-3} Å and 2,960 more have bond-length spreads within 0.05 Å.

Characteristic intersections. Species that sit at or near $z = n$, e.g., B^{3+} in trigonal coordination (B_2O_3, band gap 6 eV), C^{4+} at $n = 4$, and Si^{4+} in tetrahedral coordination (SiO_2, 9 eV), lie in the wide-gap covalent limit where weak screening is physically plausible. For these species the pole marks the breakdown of the first-order link between R_0 and B, so the B–R_0 line fitted at the pole coordination cannot by itself constrain the softness. The data, however, resolve each cation at more than one coordination number, and the n-resolved B–R_0 lines of a given cation (slopes $\beta^{(n)}$) are not independent. Plotted together, they intersect at a nearly common point (Supplemental Material [18]). Because the slopes are fixed by z/n, the intersection is set entirely by the line intercepts. Writing each n-resolved line as $B = \beta^{(n)} R_0 + \beta_0^{(n)}$, [19] the uniform-limit relation $R_0 = \bar{R} + B \ln(z/n)$ gives

$$\beta_0^{(n)} = -\frac{\bar{R}^{(n)}}{\ln(z/n)}, \tag{6}$$

where $\bar{R}^{(n)}$ is the characteristic shell radius of the species at coordination n. Each line crosses $B = 0$ at the shell radius itself, so the slope carries the charge-to-coordination ratio while the intercept records where the shell sits. The intersection of two coordination branches $n_1 < n_2$ is therefore fixed by the dilation of the shell with coordination number,

$$B^* = \frac{\bar{R}^{(n_2)} - \bar{R}^{(n_1)}}{\ln(n_2/n_1)}, \tag{7}$$

with $R_0^* = \bar{R}^{(n_1)} + B^* \ln(z/n_1)$ the common radius at which the branches agree. Bond lengths grow with coordination number, so Eq. (7) gives $B^* > 0$ naturally on the screening branch, and the characteristic softness acquires a direct geometric meaning. It is the logarithmic rate at which the coordination shell dilates with n.

These intersections define the characteristic pair (R_0^*, B^*), species-level constants fixed by the family of lines rather than by any single member, and therefore well defined even when one line steepens toward the pole. Two lines on the same branch differ in slope by $\ln(n_2/n_1)/[\ln(z/n_1)\ln(z/n_2)]$, which is small whenever z is well separated from both coordination numbers, so such lines can appear nearly coincident. Their intersection nonetheless remains constrained exactly to the extent that the data resolve the shell dilation in Eq. (7). Operationally, (R_0^*, B^*) minimizes the weighted variance of the coordination-resolved lines, with each line weighted by the number of structures retained in its final fit, and the pooled construction reports the residual spread σ_B as its conditioning diagnostic (Supplemental Material [18]). The screening-branch B^* values are not universal but cluster near the conventional $B = 0.37$ Å [2], and their comparison with published softness parameterizations [9–11] is given in the Supplemental Material [18].

On the screening branch ($\lambda > 0$, $B > 0$), inverting Eq. (4) at the shell level gives the structure-specific screening length $\lambda = (-R' + \sqrt{R'^2 + 4BR'})/2$. Evaluating the same positive-branch inversion at the characteristic point, where the operational choice of R' equals the characteristic intercept R_0^*, gives the tabulated species constant $\lambda^* = (-R_0^* + \sqrt{R_0^{*2} + 4B^* R_0^*})/2$. The root is real for $B^* \geq -R_0^*/4$. It is positive on the screening branch and negative on the anti-screening branch. The pole itself has a screening interpretation. The species at $z = n$ are wide-gap insulators (B_2O_3 and SiO_2 above), and in conventional screening theory gapped systems screen weakly because low-energy charge response is suppressed, in contrast to the finite Thomas–Fermi screening set by a metal's density of states at the Fermi level [14–16, 20]. The fitted family behaves the same way. Approaching the pole ($z = n \mp \varepsilon$, $\varepsilon \to 0$), sustaining a finite intercept offset $R_0 - R_0^*$ requires the inferred screening length to grow as $|\lambda| \sim \varepsilon^{-1/2}$, so the species where weak screening is physically expected are exactly those for which the model demands an unbounded screening length. Our coarse-grained λ is not identical to those microscopic lengths, but the two weak-screening limits agree.

The λ^* values determined from the B^* are tabulated in the Supplemental Material [18] for all 103 species, positive for the 101 screening-branch species and negative for the two anti-screening species (Mo^{3+}, Sb^{3+}), whose departures from the first-order Yukawa regime are diagnosed there. The two roots of the species-level screening length also have physical meaning. The anti-screening alias of λ^* has magnitude $|\lambda_-^*| = R_0^* + \lambda^*$. This is the characteristic shell radius extended by one screening length, the outer reach of the screened bond. Compared against Shannon crystal radii [21], $|\lambda_-^*|$ behaves

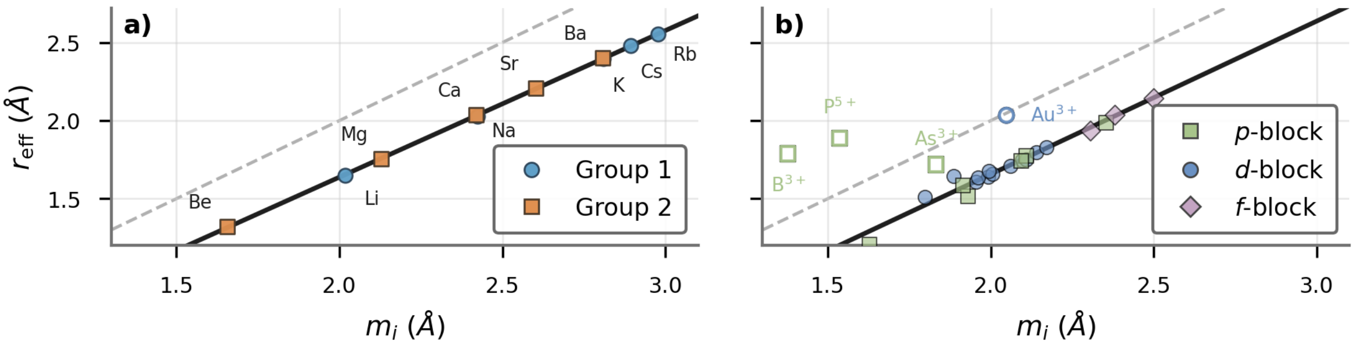

FIG. 2. First-principles screening centroid r_{eff} vs. Gibbs shell centroid m_i. (a) Ten s-block ground-state binary oxides. The solid line is the shared fit and the dashed line is $r_{\mathrm{eff}} = m_i$. (b) Twenty-five non-s-block ground-state binary oxides, color-coded by block. Open symbols mark the four directional-bonding outliers excluded from the regression (B_2O_3, P_2O_5, As_2O_3, Au_2O_3).

as a valence-dependent contact distance (Supplemental Material [18]). Here r_{cr} denotes the cation's Shannon crystal radius averaged over its observed coordination numbers, the geometric baseline for the cation–oxygen contact. The comparison is empirical, with no assumption about the physics that sets the crystallographic radii. Within each formal-charge class the slope of $|\lambda^*_-|$ against r_{cr} is fixed at one and a single offset $c(z)$ is fitted,

$$|\lambda^*_-| = r_{\mathrm{cr}} + c(z),$$

so $c(z)$ measures how far the outer reach of the screened bond extends beyond the cation radius. These one-parameter fits describe each class well, and the offsets grow linearly with formal charge, $c(z) \approx 1.05 + 0.09\, z$ Å. The two terms have direct physical readings. The constant 1.05 Å sits at the scale of the oxygen crystal radius, so at low charge $r_{\mathrm{cr}} + c(z)$ is the cation radius plus an oxygen radius, the geometric cation–oxygen contact distance. The increment shows the screened bond reaching about 0.09 Å beyond that contact per unit of formal charge, so $c(z)$ acts as an effective oxygen size. The effective oxygen size seen by the bond-valence partition is therefore not a fixed sphere. It grows with cation valence, which is the statement that the bond-valence shell follows the electronic screening cloud rather than a hard ionic contact. Comparison to first-principles electron densities tests this directly, but first requires a shell radius that the bond-valence partition itself defines.

Partition thermodynamics. Reading Eq. (2) as a statistical ensemble supplies a characteristic shell radius and also fixes the reference length R' left open above. These are the direct consequence of Eqs. (1) and (2). We introduce thermodynamic language as interpretation, and develop the structure in three steps. First, normalize the bond valences within the shell:

$$p_{ij} := \frac{S_{ij}}{\sum_k S_{ik}} = \frac{w_{ij}}{\sum_k w_{ik}} = \frac{e^{-R_{ij}/B}}{Z_i}, \tag{8}$$

where $Z_i := \sum_j e^{-R_{ij}/B}$. The middle form follows from Eq. (2), whose shell total is z_i, so p_{ij} is the normalized screened-flux weight of bond j. The last form follows from Eq. (1), in which R_0/B cancels in the ratio. The middle and last forms coincide to the first order of Eq. (3). Equation (8) is a Gibbs distribution over the coordination shell. Bond length plays the role of energy, Z_i is the canonical partition function, and B is the temperature-like scale. Second, define two coordination shell properties,

$$m_i := \sum_j p_{ij} R_{ij}, \qquad H_i := -\sum_j p_{ij} \ln p_{ij}, \tag{9}$$

the Gibbs mean—the valence-weighted centroid of the coordination shell—and the Shannon entropy of the bond-weight distribution. Third, invert Eq. (1) bond by bond, $R_0 = R_{ij} + B \ln S_{ij}$, and average over the weights p_{ij}. With $\ln S_{ij} = \ln q_i + \ln p_{ij}$, the average gives the exact identity

$$R_0 = m_i + B(\ln q_i - H_i), \tag{10}$$

where $q_i := \sum_j S_{ij}$, equal to z_i when the valence-sum rule is enforced exactly. Here H_i measures how uniformly the bond-valence weight is distributed across the coordination shell. It reaches $\ln n$ for a symmetric shell and falls below it as distortion concentrates weight on the extremal bonds. Thus, R_0 is an entropy-corrected weighted mean. It reduces to the unit-valence limit $R_0 = R_{ij}$ that Brown *et al.* [4] derived for a single bond with $S_{ij} = 1$. The Gibbs mean also fixes the reference length left open in the screened-flux derivation. We take $R' = m_i$, which centers the log-weight expansion of Eq. (3) on the weighted shell centroid that the partition function singles out and makes the operational content of R' in Eq. (4) explicit.

Empirical validation. Bond-valence parameters (R_0, B) were determined for 103 cation–oxygen species using first-principles relaxed crystal structures from the Materials Project [22, 23]. For each structure, bond strengths were obtained by solving the valence-sum and loop constraints on the cation–oxygen bond graph [19], and (R_0, B) were fitted jointly by linear least squares, not by re-optimizing B at fixed R_0. The coordination-resolved lines of Fig. 1, first introduced in Ref. [19], are then obtained by regressing B on R_0 across the per-structure pairs of each species at fixed n (at least five structures per cell), using an outlier-robust RANSAC estimator refit by least squares on its inliers. A fitted slope $\beta^{(n)}$ is accepted only with at least four inlier structures, slope uncertainty $\sigma_\beta \leq 5$, and $R^2 \geq 0.3$ [18, 24]. The resulting set spans s-, p-, d-, and f-block cations. Figure 1 shows the 94 species (150 fitted slopes) for which the prediction is finite at $n = 4$ and/or $n = 6$.

First-principles confirmation. If bond valence is captured screened flux, the shell radius resolved by the bond-valence partition—the Gibbs shell centroid m_i of Eq. (9)—should track the radius at which screening charge accumulates along each bond. We test this by comparing m_i to a first-principles screening centroid r_{eff}, the centroid of the excess electron density on the oxygen side of the bond path, extracted from charge densities in the Materials Project [22, 23]. For each M–O bond with $R_{ij} \leq 3.5$ Å, the total valence-electron density $\rho(r)$ is sampled at 200 uniformly spaced points along the nearest-image straight M–O segment. An interior density-minimum proxy on $0.15R_{ij} < r < 0.85R_{ij}$ defines the start of the oxygen-side window, and r_{eff} is then the excess-density-weighted centroid on $r_{\text{BCP}} \leq r \leq 0.93R_{ij}$ with weights $\max[\rho(r) - \rho_{\text{BCP}}, 0]$, averaged over the valid first-shell bonds of that oxide (see Supplemental Material for the exact discrete construction and fallback rule). For ten s-block binary oxides (Li_2O through BaO, Fig. 2a), these collapse onto a single line:

$$r_{\text{eff}} = 0.94\, m_i - 0.24\ \text{Å} \qquad (R^2 = 0.9998).$$

This is not a trivial monotone size trend: the jackknife mean absolute error of the m_i regression is 0.006 Å, versus 0.078 Å for a linear fit against R_0 and 0.095 Å against the Shannon cation radius [21] (see Supplemental Material for the corresponding control plot). The near-unit slope therefore shows that the bond-valence partition function and the first-principles screening cloud track the same ionic shell radius. The same centroid correspondence extends beyond the s block. Figure 2b shows the 25 non-s-block binary oxides for which the charge density yields a well-defined oxide-level screening centroid (listed in the Supplemental Material [18]). The Yukawa derivation assumes approximately isotropic first-shell screening, so the regression is restricted to the 21 oxides whose cation environments satisfy that condition, excluding four whose bonding is dominated by directional, non-isotropic motifs: B_2O_3, P_2O_5, As_2O_3, and Au_2O_3. The Supplemental Material gives the fixed inclusion rule and the exact oxide-level centroid construction used to define r_{eff}. The 21 included oxides follow

$$r_{\text{eff}} = 0.98\, m_i - 0.30\ \text{Å} \qquad (R^2 = 0.967).$$

The correspondence therefore spans the p, d, and f blocks, and its breakdown is confined to strongly anisotropic directional-bonding environments outside the isotropic Yukawa regime.

Discussion. The main result is that the bond-valence exponential is the leading-order term of Yukawa screened electrostatics. The fitted exponent denominator B is not a free parameter but the leading-order Yukawa denominator set by the screening length and shell geometry, with its sign carrying the screening/anti-screening branch information that diagnoses the applicability of the first order, linear assumption employed here. The empirical correlation between R_0 and B is determined by $\beta = 1/\ln(z/n)$ and exact in the uniform-bond limit. The partition-function structure for inequivalent bonds gives an entropy-corrected identity for R_0 and a free energy $F = m_i - BH_i = R_0 - B \ln q_i$ in length units.

The derivation rests on two approximations: (i) the first-order Taylor expansion, which breaks down for highly distorted shells (d^4/d^9 Jahn–Teller octahedra), and (ii) isotropic flux capture, in which the ansatz $w_{ij} \propto e^{-R_{ij}/\lambda}(1 + R_{ij}/\lambda)$ assigns each bond a weight that depends on its length alone. Both assumptions fail when directional bonding concentrates flux capture into particular solid angles, adding a bond-dependent geometric term that the expansion of Eq. (3) cannot absorb into C_i, as in stereoactive lone-pair and strongly anisotropic environments [4, 25]. The empirical signature of this breakdown is visible in Fig. 2b: the four oxides whose ground-state structures fail the isotropic flux-capture assumption (B_2O_3, P_2O_5, As_2O_3, Au_2O_3) are precisely those that depart from the centroid line, identifying ground-state directional bonding as the empirical boundary of the first-order Yukawa regime. Extending the first-principles centroid comparison to the full set of transition-metal and f-block oxides beyond the subset treated here will require going beyond the Thomas–Fermi approximation to include orbital directionality and correlation effects.

The first-principles centroid comparison confirms that this is not merely a mathematical analogy, but that the electronic screening cloud and the bond-valence weights describe the same shell geometry. The offset from the identity line has a direct origin. The Gibbs shell centroid m_i is built from internuclear M–O distances and therefore references the oxygen nucleus, whereas r_{eff} tracks the oxygen-side screening charge, whose centroid is displaced from the nucleus into the bond. The same charge-density paths measure this oxygen-side valence shift directly, $\Delta_{\text{O,val}} = 0.39 \pm 0.03$ Å on average, matching the $m_i - r_{\text{eff}}$ gap of the fitted lines across the shell radii (Supplemental Material [18]). The null-model controls show that the collapse is substantially sharper than a generic monotone size trend.

Bond valence is therefore recast as a coarse-grained observable of local screened charge response in specific screening environments. Through species-level aggregation of screening across formal charge and coordination number, the screened bond coincides with the cation's Shannon crystal radius plus an effective oxygen size that grows with cation valence. Establishing the full predictive scope of that program beyond the families validated here remains future work. This connects Pauling's 1929 valence rules [1] to Yukawa's 1935 screened potential [17] through a quantitative bridge that has been missing for nearly a century.

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